

BITS PILANI, DUBAI CAMPUS
DUBAI INTERNATIONAL ACADEMIC CITY, DUBAI

COURSE	: COMPUTER PROGRAMMING (CS F111)
COMPONENT	: Practical Week 9: String handling

Q.1 Write a C program to accept first name, middle name and last name from user, store them in 3 different character arrays. Concatenate these 3 strings in one and print as full name.

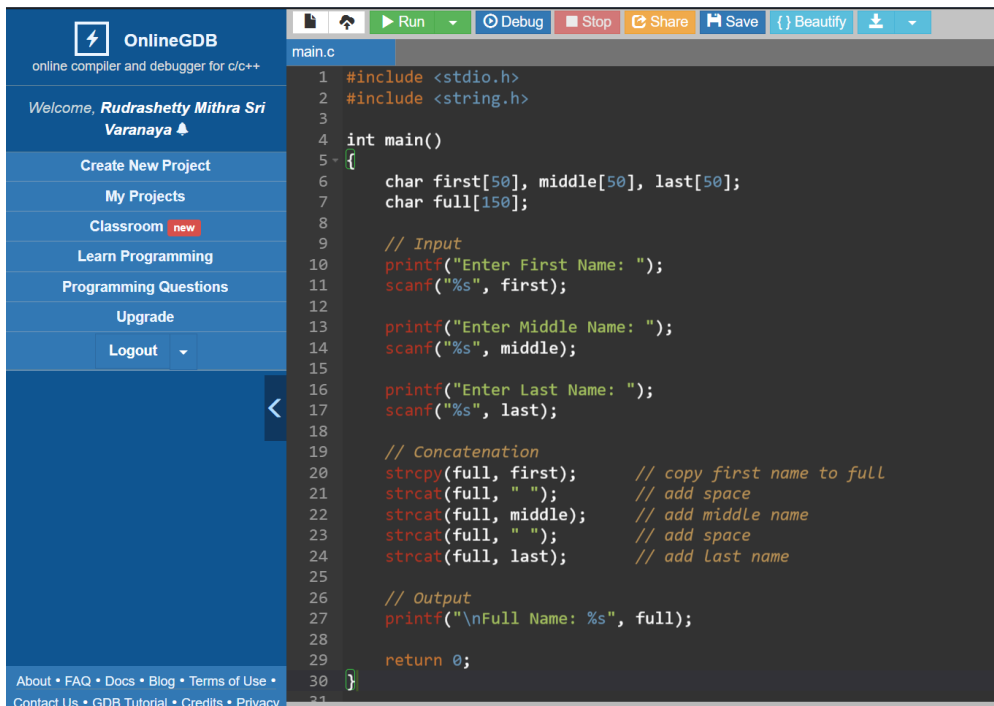
Test case

char first[20], char middle[20], char last[20];

gets(first); similarly take input for middle and last

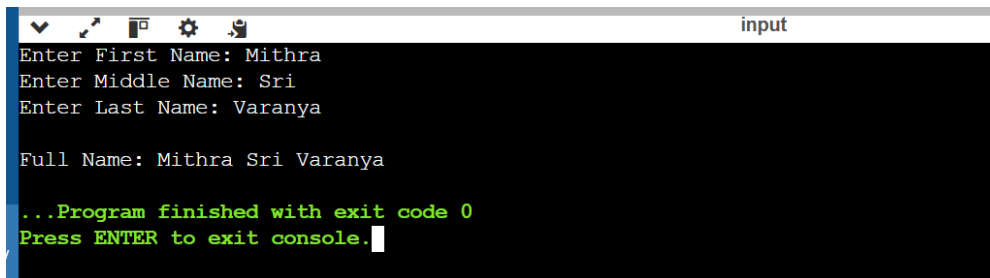
strcat(first,middle); // first= first+middle, similarly do for last

Answer:code:



```
1 #include <stdio.h>
2 #include <string.h>
3
4 int main()
5 {
6     char first[50], middle[50], last[50];
7     char full[150];
8
9     // Input
10    printf("Enter First Name: ");
11    scanf("%s", first);
12
13    printf("Enter Middle Name: ");
14    scanf("%s", middle);
15
16    printf("Enter Last Name: ");
17    scanf("%s", last);
18
19    // Concatenation
20    strcpy(full, first);    // copy first name to full
21    strcat(full, " ");    // add space
22    strcat(full, middle);    // add middle name
23    strcat(full, " ");    // add space
24    strcat(full, last);    // add last name
25
26    // Output
27    printf("\nFull Name: %s", full);
28
29    return 0;
30 }
```

Input and output:



```
input
Enter First Name: Mithra
Enter Middle Name: Sri
Enter Last Name: Varanya

Full Name: Mithra Sri Varanya

...Program finished with exit code 0
Press ENTER to exit console.
```

Q.2 Write a C Program to get two string inputs from the user. Compare the two strings using string function and print strings are equal or not.

Test Case

Input 2 strings using gets(S1);

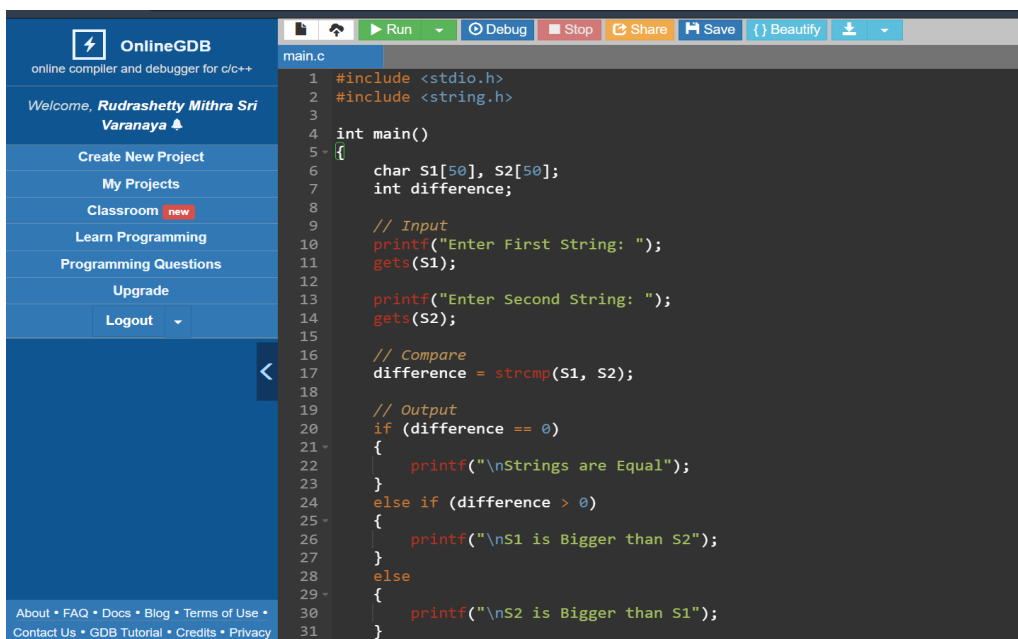
Int difference=strcmp(S1,S2);

If difference =0 then Strings are same

Else If difference >0 then S1 is bigger than S2

Else If difference <0 then S2 is bigger than S1

Answer: code:

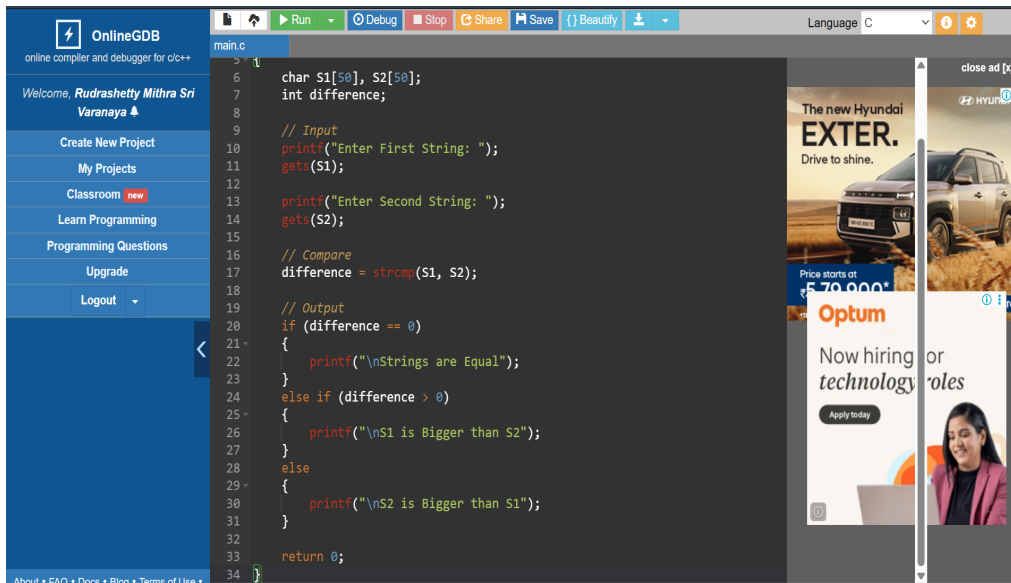


```
OnlineGDB
online compiler and debugger for c/c++

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main.c
1 #include <stdio.h>
2 #include <string.h>
3
4 int main()
5 {
6     char S1[50], S2[50];
7     int difference;
8
9     // Input
10    printf("Enter First String: ");
11    gets(S1);
12
13    printf("Enter Second String: ");
14    gets(S2);
15
16    // Compare
17    difference = strcmp(S1, S2);
18
19    // Output
20    if (difference == 0)
21    {
22        printf("\nStrings are Equal");
23    }
24    else if (difference > 0)
25    {
26        printf("\nS1 is Bigger than S2");
27    }
28    else
29    {
30        printf("\nS2 is Bigger than S1");
31    }
```



Input and outputs:

Case1: Equal

```
input
/nix/store/7h3qngvkv6z2r8lq4j5mks4l6r5x2cq-binutils-2.44/bin/ld: /tmp/ccPWAYJU.o: in function `main':
main.c:(.text+0x24): warning: the `gets' function is dangerous and should not be used.
Enter First String: apple
Enter Second String: apple

Strings are Equal

...Program finished with exit code 0
Press ENTER to exit console.
```

Case2: S1 bigger

```
input
/nix/store/7h3qngvkv6z2r8lq4j5mks4l6r5x2cq-binutils-2.44/bin/ld: /tmp/ccLWUWOE.o: in function `main':
main.c:(.text+0x24): warning: the `gets' function is dangerous and should not be used.
Enter First String: mango
Enter Second String: apple

S1 is Bigger than S2

...Program finished with exit code 0
Press ENTER to exit console.
```

Case3: S2 bigger

```
input
/nix/store/7h3qngvkv6z2r8lq4j5mks4l6r5x2cq-binutils-2.44/bin/ld: /tmp/ccqBKOZm.o: in function `main':
main.c:(.text+0x24): warning: the `gets' function is dangerous and should not be used.
Enter First String: ant
Enter Second String: bat

S2 is Bigger than S1

...Program finished with exit code 0
Press ENTER to exit console.
```

Q.3 Write a C program to accept a string from the user and count the number of vowels and consonants in it.

Test case

char s[]="Computer lab"

if(s[i]=='a' || s[i]=='A's[i]=='u' || s[i]=='U')

vowel++;

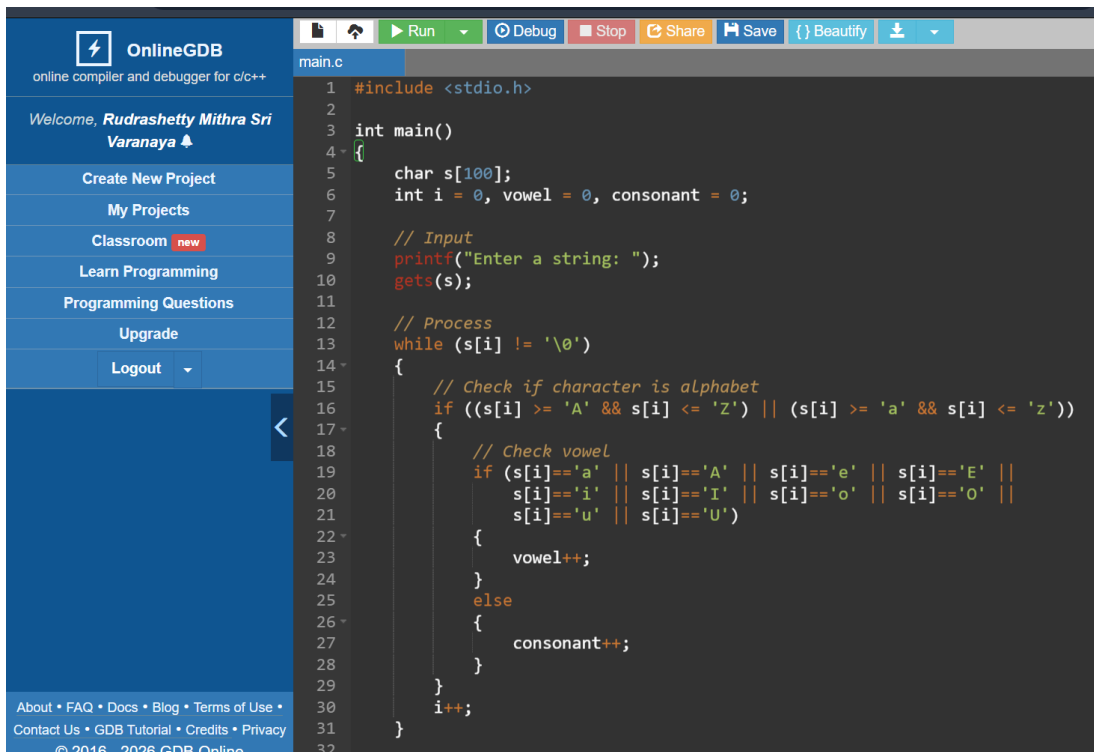
else

consonant++;

Vowels : o, u, e, a = 4

Consonants: C, m, p, t, r, l, b = 7

Answer: Code:



```
1  #include <stdio.h>
2
3  int main()
4  {
5      char s[100];
6      int i = 0, vowel = 0, consonant = 0;
7
8      // Input
9      printf("Enter a string: ");
10     gets(s);
11
12     // Process
13     while (s[i] != '\0')
14     {
15         // Check if character is alphabet
16         if ((s[i] >= 'A' && s[i] <= 'Z') || (s[i] >= 'a' && s[i] <= 'z'))
17         {
18             // Check vowel
19             if (s[i]=='a' || s[i]=='A' || s[i]=='e' || s[i]=='E' ||
20                 s[i]=='i' || s[i]=='I' || s[i]=='o' || s[i]=='O' ||
21                 s[i]=='u' || s[i]=='U')
22             {
23                 vowel++;
24             }
25             else
26             {
27                 consonant++;
28             }
29         }
30         i++;
31     }
32 }
```

The screenshot shows the OnlineGDB web interface. On the left is a sidebar with navigation links: 'Welcome, Rudrashetty Mithra Sri Varanaya', 'Create New Project', 'My Projects', 'Classroom' (marked as new), 'Learn Programming', 'Programming Questions', 'Upgrade', and 'Logout'. The main area displays a C program in a dark-themed editor. The program prompts the user to enter a string and counts the number of vowels and consonants. The code is as follows:

```
7
8 // Input
9 printf("Enter a string: ");
10 gets(s);
11
12 // Process
13 while (s[i] != '\0')
14 {
15     // Check if character is alphabet
16     if ((s[i] >= 'A' && s[i] <= 'Z') || (s[i] >= 'a' && s[i] <= 'z'))
17     {
18         // Check vowel
19         if (s[i]=='a' || s[i]=='A' || s[i]=='e' || s[i]=='E' ||
20             s[i]=='i' || s[i]=='I' || s[i]=='o' || s[i]=='O' ||
21             s[i]=='u' || s[i]=='U')
22         {
23             vowel++;
24         }
25         else
26         {
27             consonant++;
28         }
29     }
30     i++;
31 }
32
33 // Output
34 printf("\nVowels: %d", vowel);
35 printf("\nConsonants: %d", consonant);
36
37 return 0;
38 }
```

At the bottom of the sidebar, there are links for 'About', 'FAQ', 'Docs', 'Blog', 'Terms of Use', 'Contact Us', 'GDB Tutorial', 'Credits', and 'Privacy'. The copyright notice at the bottom reads '© 2016 - 2026 GDB Online'.

Input and Output:

The screenshot shows the output of the program in a terminal window titled 'input'. The output text is as follows:

```
/nix/store/7h3qngvkv6z2r8lq4j5mks4l6r5x2cq-binutils-2.44/bin/ld: /tmp/ccHPDZlE.o: in function `main':
main.c:(.text+0x39): warning: the `gets' function is dangerous and should not be used.
Enter a string: computer lab

Vowels: 4
Consonants: 7

...Program finished with exit code 0
Press ENTER to exit console.
```